



ANDERSON JUNIOR COLLEGE

Preliminary Examinations 2012

CHEMISTRY

Higher 1

Paper 1 Multiple Choice

8872/01

19 September 2012

50 minutes

Additional Materials: Multiple Choice Answer Sheet
 Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the Multiple Choice Answer Sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

Multiple Choice Answer Sheet

Write your name, PDG and NRIC/FIN number, including the reference letter.

Shade the NRIC / FIN number.

Exam Title: JC2 Prelim

Exam Details: H1 Chem / Paper 1

Date: 19/09/2012

This document consists of **13** printed pages.

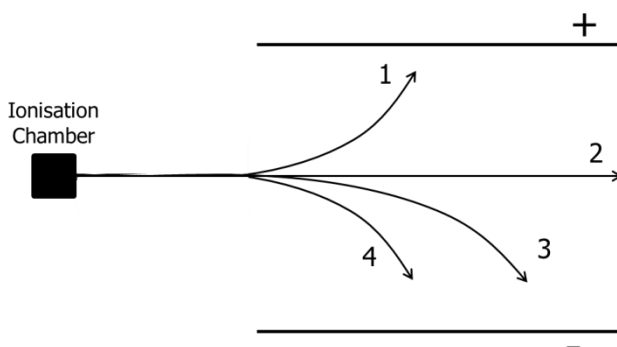
Section A

For each question there are four possible answers, **A**, **B**, **C** and **D**. Choose the **one** you consider to be correct.

- 1 Americium–241, commonly used in smoke detectors, undergoes decay by emitting a constant stream of alpha particles which are similar to the nucleus of ^4He .

Strontium–90 is another radioactive substance which can be used as a tracer for medicinal or agriculture uses. On decaying, it emits beta particles which can be considered as electrons.

A small amount of Americium–241 and Strontium–90 is separately placed in an ionisation chamber to emit a constant stream of radiation and the emitted particles are passed through an electric field.

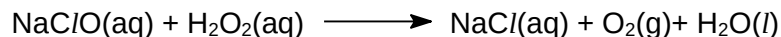


What would be the path of the emitted particles in an electric field?

	^{241}Am (α -particles)	^{90}Sr (β -particles)
A	1	2
B	1	3
C	4	1
D	3	1

- 2 Which substance, in 1.00 mol dm^{-3} aqueous solution, would have the same hydrogen ion concentration as 1.00 mol dm^{-3} of hydrochloric acid?
- A Ethanoic acid
- B Nitric acid
- C Sodium hydroxide
- D Sulfuric acid

- 3** A household bleach contains sodium chlorate(I), NaClO , as its active ingredient. The concentration of NaClO in the bleach can be determined by reacting a known amount with aqueous hydrogen peroxide, H_2O_2 .



When 25.0 cm³ of bleach is treated with an excess of aqueous H₂O₂, approximately 840 cm³ of oxygen gas, measured at room temperature and pressure, is given off.

What is the concentration of NaClO in the bleach?

- A** $8.75 \times 10^{-4} \text{ mol dm}^{-3}$
B $0.700 \text{ mol dm}^{-3}$
C $0.875 \text{ mol dm}^{-3}$
D 1.40 mol dm^{-3}

- 4** The successive ionisation energies, in kJ mol^{-1} , of an element **E** are given below.

870 1800 3000 3600 5800 7000 13200

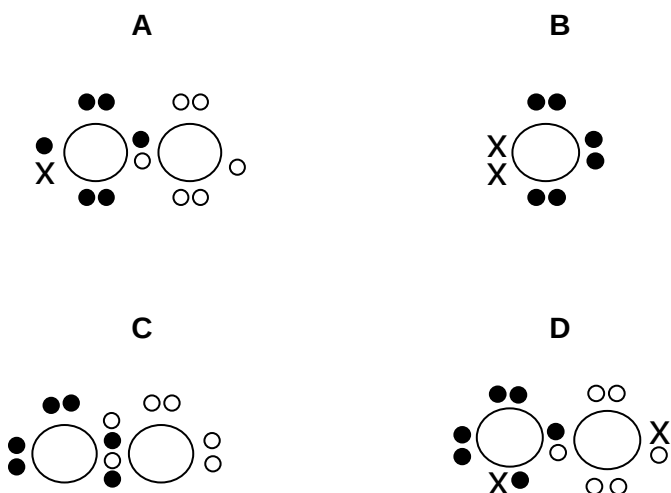
Which of the following could be **E**?

- A** As **B** Zr **C** Te **D** I

- 5** Lithium peroxide, Li_2O_2 , is synthesized by reacting lithium hydroxide with hydrogen peroxide.

Which diagram correctly shows the bonding of the peroxide anion in Li_2O_2 ?

Key: ● electron from the first oxygen atom
○ electron from the second oxygen atom
× electron from lithium atom



- 6 Methyl isocyanate, $\text{CH}_3\text{-N=C=O}$, is a toxic liquid which is used in the manufacture of some pesticides.

What is the approximate angle between the bonds formed by the N atom?

- A 104° B 109° C 120° D 180°

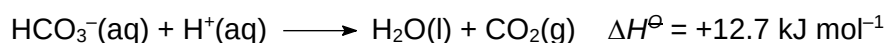
- 7 Three elements, **F**, **G** and **H**, have the physical properties shown in the table.

Element	Melting point / $^\circ\text{C}$	Boiling point / $^\circ\text{C}$	Density / g cm^{-3}
F	1400	3300	2.33
G	97	883	0.97
H	-7	59	3.12

What could be the identities of **F**, **G** and **H**?

	F	G	H
A	silicon	sodium	bromine
B	magnesium	aluminium	iodine
C	sodium	magnesium	bromine
D	aluminium	silicon	iodine

- 8 Solutions of hydrogencarbonate can react with acids as follows.



Given the following appropriate enthalpy changes of formation, what is the enthalpy of formation of $\text{H}^+(\text{aq})$?

Species	$\Delta H_f^\ominus / \text{kJ mol}^{-1}$
$\text{H}_2\text{O}(\text{l})$	-285.8
$\text{CO}_2(\text{g})$	-393.5
$\text{HCO}_3^-(\text{aq})$	-692.0

- A $-25.4 \text{ kJ mol}^{-1}$
 B 0 kJ mol^{-1}
 C $+25.4 \text{ kJ mol}^{-1}$
 D $+1384 \text{ kJ mol}^{-1}$

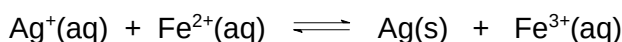
- 9 50 cm³ of 2.5 mol dm⁻³ hydrochloric acid was placed in a polystyrene beaker of negligible heat capacity. Its temperature was recorded and then 50 cm³ of 2.5 mol dm⁻³ NaOH at the same temperature was quickly added, with stirring. The temperature rose by 17°C.

The resulting solution may be considered to have a heat capacity per unit volume of 4.2 J cm⁻³ K⁻¹.

What is an approximate value for the molar enthalpy change of neutralisation of hydrochloric acid and sodium hydroxide from this experiment?

- A $\frac{-(50 \times 4.2 \times 17)}{(0.050 \times 2.5)} \text{ J mol}^{-1}$
- B $\frac{-(50 \times 4.2 \times 17)}{(0.10 \times 2.5)} \text{ J mol}^{-1}$
- C $\frac{-(100 \times 4.2 \times 17)}{(0.050 \times 2.5)} \text{ J mol}^{-1}$
- D $\frac{-(100 \times 4.2 \times 17)}{(50 \times 2.5)} \text{ J mol}^{-1}$

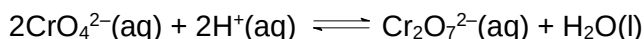
- 10 An aqueous solution, containing 1.0 mol of AgNO₃ and 1.0 mol of FeSO₄ in 2.00 dm³ of water, was prepared and allowed to reach equilibrium. When equilibrium was established, there was 0.44 mol of Ag⁺(aq) in the mixture.



What is the numerical value of K_c ?

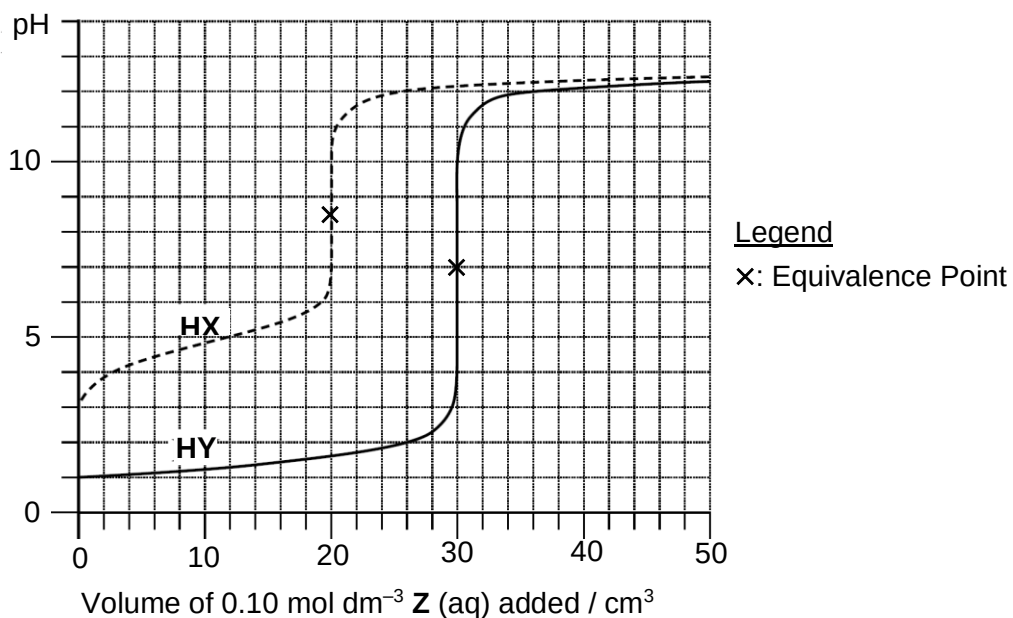
- A 1.62 B 2.06 C 2.89 D 5.79

- 11 Which of the statements concerning the equilibrium given below is true?



- A An increase in pH will increase the [Cr₂O₇²⁻(aq)].
- B There is a change in the oxidation state of chromium.
- C The equilibrium constant, K_c , has the units of mol⁻² dm⁶.
- D Diluting the system will cause a shift in the position of equilibrium to the left.

- 12 The titration curves below were obtained using equal volumes of solutions of two monobasic acids **HX** and **HY**.

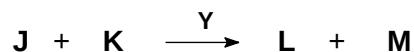


Which of the following sets of reagents produces the above data?

	HX	HY	Z	[HX]/[HY]
A	CH ₃ COOH	HCl	NH ₃	2/3
B	CH ₃ COOH	HCl	NaOH	2/3
C	HCl	CH ₃ COOH	NH ₃	3/2
D	HCl	CH ₃ COOH	NaOH	3/2

- 13 Which of the following statements about the rate constant, k , of chemical reactions is **not** true?
- A** The rate constant increases when a catalyst is added.
 - B** The rate constant increases when the temperature is increased.
 - C** The rate constant increases when the concentration of reactants is increased.
 - D** The rate constant can have different units.

- 14** Using a colorimeter, the following reaction is studied by finding the time taken for the colored reactant, **J**, to decolourise. The reaction is catalysed by **Y**.



In this method, the rate of reaction is measured in terms of the rate at which the concentration of **J** changes

$$\text{i.e. rate of reaction} \propto \frac{\text{concentration of J added}}{\text{time taken for colour of J to disappear}}$$

The following results were obtained.

Experiment	Volume of J added / cm ³	Volume of K added / cm ³	Volume of Y added / cm ³	Volume of H ₂ O added / cm ³	Time taken for colour of J to disappear / s
1	5	5	5	10	40
2	5	20	10	15	10
3	10	20	10	10	20
4	10	20	5	15	40

- A** Rate = $k [\text{J}] [\text{Y}]$
B Rate = $k [\text{K}] [\text{Y}]$
C Rate = $k [\text{J}] [\text{K}]$
D Rate = $k [\text{J}] [\text{K}] [\text{Y}]$

- 15** Consecutive elements **P**, **Q** and **R** are in the third period of the Periodic Table. Element **Q** has the highest first ionisation energy and the lowest melting point.

What could be the identities of **P**, **Q** and **R**?

- A** Silicon, phosphorus, sulfur
B Aluminium, silicon, phosphorus
C Magnesium, aluminium, silicon
D Sodium, magnesium, aluminium

- 16** The chloride of element **S** neither reacts with nor dissolves in water.

Which element could **S** be?

- A** Aluminium
B Carbon
C Magnesium
D Silicon

- 17 Two properties of non-metallic elements and their atoms are as follows.

Property 1 : the oxide can form a strong acid in water

Property 2 : the atom has no paired 3p electrons

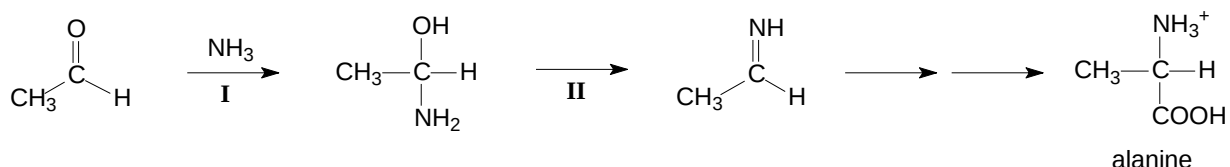
Which properties do phosphorus and sulfur have?

	Phosphorus	Sulfur
A	1 and 2	1 only
B	1 only	1 and 2
C	1 and 2	1 and 2
D	2 only	1 only

- 18 Which compound could **not** be obtained from cracking a sample of nonane, $\text{CH}_3(\text{CH}_2)_7\text{CH}_3$?

- A $(\text{CH}_3\text{CH}_2\text{CH}_2)_3\text{CH}$
 B $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_3$
 C $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}=\text{CH}_2$
 D $\text{CH}_3\text{CH}=\text{CHCH}=\text{CHCH}_2\text{CH}_2\text{CH}_2\text{CH}_3$

- 19 The following reaction scheme shows the synthesis of an amino acid, alanine.



Which of the following shows the correct types of reaction for steps I and II?

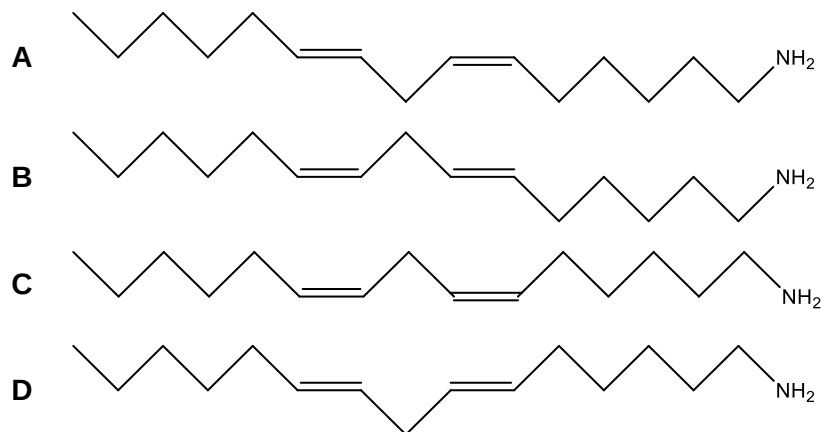
	I	II
A	Substitution	Elimination
B	Substitution	Condensation
C	Addition	Elimination
D	Addition	Addition

- 20 When 0.1 mol of a hydrocarbon **T** is burnt in excess oxygen, 17.6 g of carbon dioxide is produced. **T** is also found to react with chlorine gas to form hydrogen chloride as a by-product.

What is the possible structural formula of **T**?

- A $\text{CH}_3\text{CH}_2\text{CH}_3$
 B $\text{CH}_3\text{CH}=\text{CH}_2$
 C $\text{CH}_3\text{CH}=\text{CHCH}_3$
 D $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$

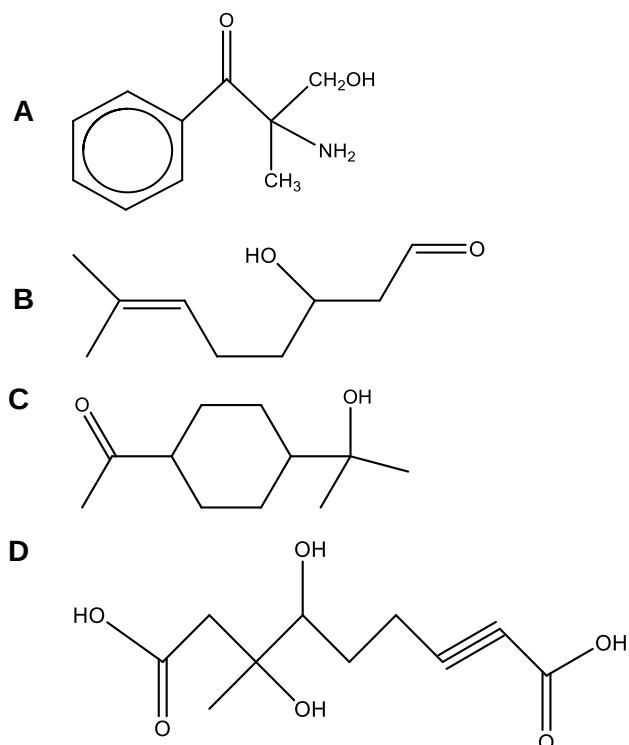
21 Which of the following formulae contains a *trans-trans* arrangement?



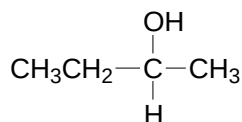
22 An organic compound **U** has the following properties:

- it gives an orange precipitate with 2,4-dinitrophenylhydrazine;
- it does not react with aqueous diamminesilver ions;
- it produces fumes of HCl gas with PCl₅;
- it turns orange acidified K₂Cr₂O₇ green.

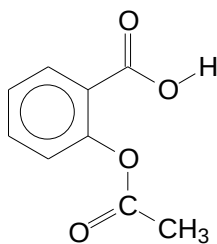
Which one of the following could **U** be?



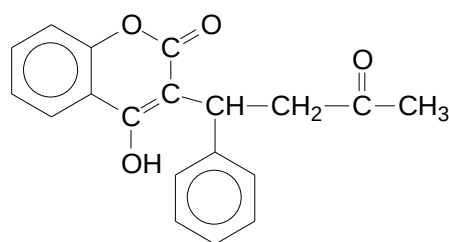
- 23 The structures of butan-2-ol, aspirin and warfarin are shown.



butan-2-ol



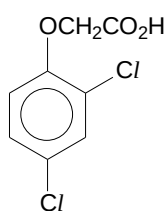
aspirin



warfarin

Which of the above organic compound(s) on warming with alkaline aqueous iodine produces a yellow precipitate?

- A butan-2-ol only
 B butan-2-ol and aspirin only
 C butan-2-ol and warfarin only
 D butan-2-ol, aspirin and warfarin
- 24 The compound 2,4-D is used as a weedkiller.

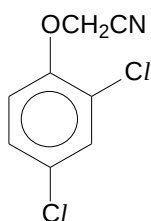


2,4-D

Which of the following statements about this compound is **incorrect**?

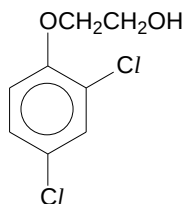
- A It dissolves in $\text{Na}_2\text{CO}_3(\text{aq})$, liberating $\text{CO}_2(\text{g})$.
 B It can be esterified by an alcohol in the presence of H^+ ions.

- C It can be obtained from

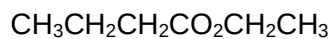


on heating with dilute H_2SO_4 .

- D It reacts with NaBH_4 to form



- 25 An ester with an odour of pineapple has the following structure.



In which of the following will the substances react together under suitable conditions to produce this ester?

- A $\text{CH}_3\text{CH}_2\text{CH}_2\text{CO}_2\text{H} + \text{CH}_3\text{CH}_2\text{OH}$
- B $\text{CH}_3\text{CH}_2\text{CH}_2\text{CO}_2\text{H} + \text{CH}_3\text{CH}_2\text{Cl}$
- C $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{OH} + \text{CH}_3\text{CO}_2\text{H}$
- D $\text{CH}_3\text{CH}_2\text{CH}_2\text{CO}_2\text{H} + \text{CH}_3\text{CO}_2\text{H}$

Section B

For each of the question in this section, one or more of the three numbered statements **1** to **3** may be correct.

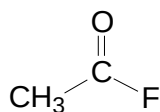
Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct.)

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

26 Which types of intermolecular forces can exist between adjacent acetyl fluoride molecules?



acetyl fluoride

- 1** hydrogen bonding
 - 2** temporary dipole–induced dipole forces
 - 3** permanent dipole–dipole forces
- 27** Which set of solutions of equal volume when mixed will **not** give an acidic buffer?
- 1** 0.10 mol dm⁻³ CH₃CO₂H and 0.10 mol dm⁻³ NaOH
 - 2** 0.10 mol dm⁻³ CH₃CO₂H and 0.05 mol dm⁻³ Ca(OH)₂
 - 3** 0.05 mol dm⁻³ HCl and 0.05 mol dm⁻³ CH₃CO₂Na
- 28** Which statement concerning only the elements in the third period, sodium to argon is correct?
- 1** The element with the highest electrical conductivity is aluminium.
 - 2** The element that has exactly four atoms in its molecule is phosphorus.
 - 3** The one with the highest melting point is silicon.

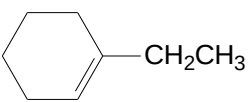
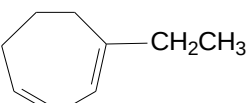
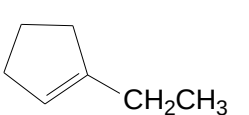
The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

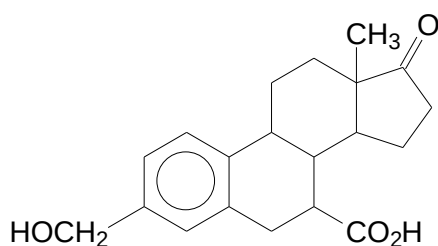
No other combination of statements is used as a correct response.

- 29** A hydrocarbon, on heating with an excess of concentrated acidified $\text{KMnO}_4(\text{aq})$, produces $\text{HO}_2\text{C}(\text{CH}_2)_3\text{COCH}_2\text{CH}_3$.

What could the hydrocarbon be?

- 1** 
- 2** 
- 3** 

- 30** Which of the following statements is true about 1 mol of the molecule shown below?



- 1** It gives an orange precipitate with 2,4-dinitrophenylhydrazine.
- 2** It produces 1 mol of hydrogen gas on reaction with excess sodium metal.
- 3** It forms an ester with ethanoic acid in the presence of dilute sulfuric acid.

<i>Question Number</i>	<i>Key</i>	<i>Question Number</i>	<i>Key</i>
1	D	16	B
2	B	17	A
3	D	18	A
4	C	19	C
5	D	20	D
6	C	21	D
7	A	22	A
8	B	23	C
9	C	24	D
10	D	25	A
11	D	26	C
12	B	27	A
13	C	28	A
14	B	29	C
15	A	30	B